A Termwise Proof of an OEIS Supercongruence

Prime-indexed binomial sums in A365029

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Abstract. OEIS A365029 conjectures that a particular binomial sum satisfies $a(p-1) \equiv a(0) \pmod{p^3}$ for every prime $p \geq 5$. We give a termwise proof that works for every odd prime, and obtain a more general congruence of modulus p^{A+B} for a family of exponents. The other, iterative supercongruence in the same entry is not addressed.

Exact OEIS assertion

The sequence A365029 is defined by

$$a(n) = \sum_{k=0}^{n} \binom{n+k-1}{k}^2 \binom{2k-1}{n}, \qquad n \geq 0, \tag{1}$$

with generalized binomial coefficients when the upper argument is negative. Its entry states two conjectured supercongruences for primes $p \geq 5$, of which the first is

$$a(p-1) \equiv a(0) = 1 \pmod{p^3}. \tag{2}$$

The second compares $a(np^r)$ with $a(np^{r-1})$ and is separate from (2).

Theorem 1 (Termwise prime-power divisibility)**.** *For integers $A, B \geq 1$ put*

$$S_{A,B}(n) = \sum_{k=0}^{n} \binom{n+k-1}{k}^A \binom{2k-1}{n}^B.$$

For every odd prime p,

$$\boxed{S_{A,B}(p-1) \equiv 1 \pmod{p^{A+B}}.}$$

In particular $a(p-1) \equiv 1 \pmod{p^3}$ for every odd prime p, proving (2).

Proof. Set $n = p-1$, which is even. The $k = 0$ summand equals

$$\binom{p-2}{0}^A \binom{-1}{p-1}^B = 1^A((-1)^{p-1})^B = 1.$$

We show that each summand with $1 \leq k \leq p-1$ is divisible by p^{A+B} or is zero.

If $1 \leq k \leq (p-1)/2$, then $2k-1 < p-1$. Thus the ordinary binomial coefficient $\binom{2k-1}{p-1}$ is zero, and so is the summand.

For the remaining indices $(p+1)/2 \leq k \leq p-1$, consider the two integer binomial coefficients

$$U_k = \binom{p+k-2}{k}, \qquad V_k = \binom{2k-1}{p-1}.$$

In the factorial-product expression

$$U_k = \frac{(p-1)p(p+1)\cdots(p+k-2)}{k!},$$

the numerator contains p, while $k!$ is prime to p since $k < p$. Hence $p \mid U_k$. Likewise,

$$V_k = \frac{(2k-p+1)(2k-p+2)\cdots(2k-1)}{(p-1)!}.$$

The numerator interval contains p: its lower endpoint satisfies $2k-p+1 \le p-1$, and its upper endpoint is at least p. Since $(p-1)!$ is prime to p, we obtain $p \mid V_k$. Consequently $p^{A+B} \mid U_k^A V_k^B$ for every index in this range.

Every nonzero summand except $k = 0$ is therefore a multiple of p^{A+B}. Summing proves the congruence. □

Checks, scope, and provenance

For $p = 3, 5, 7$, the sequence gives respectively

$$a(2) = 28 \equiv 1 \pmod{27}, \qquad a(4) = 44876 \equiv 1 \pmod{125}, \qquad a(6) = 104056597 \equiv 1 \pmod{343}.$$

The argument proves a precise OEIS assertion and a wider family, but *does not* establish the conjecture $a(np^r) \equiv a(np^{r-1}) \pmod{p^{3r}}$. No claim of novelty or priority is made; this is an independent, elementary explanation of the stated assertion.

AI authorship and verification disclosure. This note was prepared by ChatGPT (OpenAI). Algebraic reasoning and sample values were checked computationally, and the PDF was rendered and inspected. There has been no independent human peer review.

References

[1] OEIS Foundation, *A365029: Sum of squared binomial coefficients weighted by another binomial coefficient*, https://oeis.org/A365029, accessed 14 November 2026.